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Backlash Effect on Dynamic Analysis of a Two-Stage Spur Gear System

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Gearbox dynamics are characterized by a periodically changing stiffness due to multiple teeth contacts. In real gear systems, a backlash also exists that can lead to a loss in contact between the teeth. Due to this loss of contact, the gear has piecewise linear stiffness characteristics. This paper examines the effect of backlash in the two-stage gear system. A purely torsional gear system is formed by three shafts connected to each other by two spur gear pairs. Using standard methods for nonlinear systems (Newton-Raphson algorithm), the dynamic behavior of a gear system with backlash is examined. Amplitude jumps in systems due to backlash are observed.

Keywords: backlash, gearmesh stiffness fluctuation, two-stage gear system

Introduction

Several studies have been done to describe the dynamic behavior of nonlinear mechanical systems under external excitations.[1,2] These studies are necessary because a linear model is often insufficient to correctly describe dynamic behavior, and models that include nonlinear structures are able to describe the real dynamic behavior.[3] Among the structures requiring nonlinear models, gearings are increasingly characterized locally by nonlinearities.[4-7]

Gearing systems are used to reduce the rotational speed, increase the available torque, change the direction of the power transmission, and distribute the available power between several machines.

Gearing reducers, in various configurations and covering a range of powers, are considered. A decision choice involves finding a configuration that minimizes the acoustic broadcasts and determines the origins of the acoustic emissions. Numerous studies have been done that examine the dynamic behavior of a single-spur gear system,[8,9] making it a logical starting point for this investigation.

These studies have examined tooth loads and natural frequencies with linear, time-invariant models. Litak and Friswell[10] examined mesh stiffness changes in the context of a static analysis and also studied the dynamic response of a purely rotational model. Saada and Velex[11] used a linear model similar to that of Kahraman and Singh[4] to demonstrate the dominant role of mesh stiffness in gear noise. Kahraman and Singh[4] proposed a rotary version of this impact pair model. In the mathematical model, elastic teeth contact is modeled by a spring. In practice, this should be a time-varying problem because of the changing number of teeth in contact at a given mesh as the gears rotate. Furthermore, the problem is nonlinear because of the contact loss typical of resonant operation. Therefore, the mesh stiffness varies with time due to multiple teeth contacts, and the backlash between the teeth gives rise to complex behavior. Mesh stiffness variation is a prominent excitation source that may cause instabilities in gears. Contact loss occurs with large amplitude vibrations, and a typical gear system is a nonlinear oscillator.[3,6,8]

Models by Kahraman and Blankenship[1] included the gear backlash and mesh stiffness fluctuations and determined system behavior for a single gear pair system. A strong contribution to the behavior of systems with several degrees of freedom, as in multistage gear transmissions, can be modeled to help ensure a minimal reduction ratio and a maximal couple.[12-15] Recently, Kahraman and Al-Shyyab[6] proposed a nonlinear time-varying dynamic model of a two-stage gear system. The system was reduced to a two-degrees-of-freedom model by using the relative gear mesh displacements. Dimensionless

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equations of motion were solved by using a harmonic balance method.

This paper focuses on the nonlinear dynamic behavior of a two-stage gear reducer and integrates into that behavior the mesh stiffness fluctuation and the phenomenon of contact loss during working.[6,8,10] An example gear train was used to investigate the influence of gear mesh damping, backlash amplitude, and time step.

Nonlinear Model of the Two-Stage Spur Gear System

In this paper, the dynamic behavior of a two-stage gear reducer is analyzed while taking into account teeth flexibility and nonlinear behavior resulting from contact loss during service.[6,8,10] The physical model of the studied system is presented in Fig. 1.

The reducer elements are modeled by lumped parameters. The gears are modeled by concentrated masses,[13,15] and the rigidities are modeled by nonlinear springs with fluctuated stiffness and a linear damping element. At this level, the shafts and bearings are considered to be rigid. The dynamic model associated with the physical model is represented in Fig. 2 and is characterized by two degrees of freedom that are the result of teeth deflections at every stage.

Every deflection represents the relative displacement of the teeth projected by the lines of action and can be expressed by:[16]

$$\delta_1(t) = r_1\theta_1(t) + r_2\theta_2(t) \text{ and } \delta_2(t) = r_3\theta_3(t) + r_4\theta_4(t) \qquad \text{(Eq 1)}$$

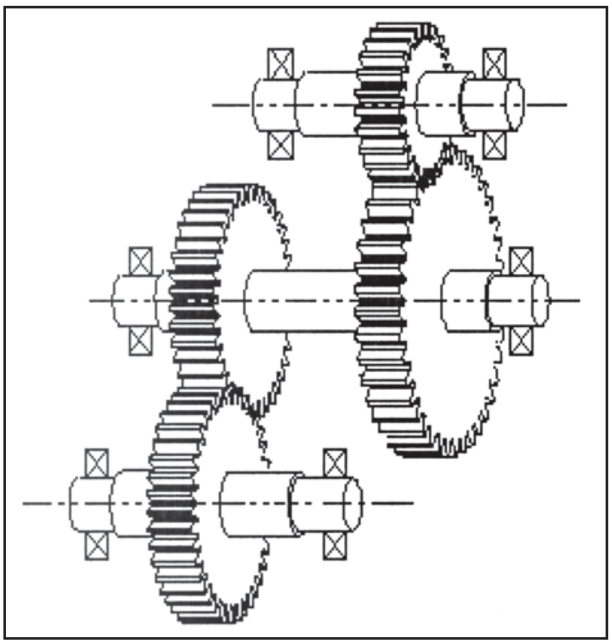

Fig. 1 *Physical model of the two-stage gear system*

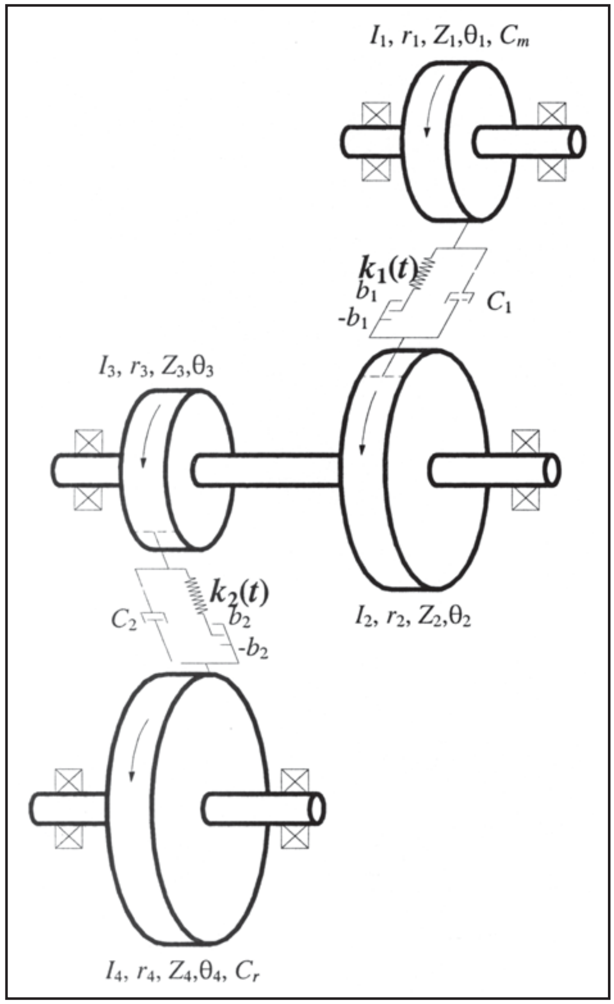

Fig. 2 *Nonlinear dynamic model of the two-stage gear system*

Table 1 Parameters of the Studied System

Material: 42CrMo4	Motor Torque	Pressure Angle	Average Mesh Stiffness	Teeth Module	Teeth Width
ρ = 7860 Kg/m^3	Cm = 1000 Nm	α = 20°	k_{moy} = 10^8 N/m	m = 410^{-3}	b = 10^{-3} m

Teeth Number		Contact Ratio	
First Gearmesh Contact	Second Gearmesh Contact	First Gearmesh Contact	Second Gearmesh Contact
Z_1 = 25; Z_2 = 40	Z_3 = 30; Z_4 = 35	$\varepsilon_{\alpha 1}$ = 1.5	$\varepsilon_{\alpha 2}$ = 1.6

where $\theta_j(t)$ represents the angular displacement of the gear j. Table 1 regroups the parameters for the two-stage gear system that was analyzed.

Excitation Sources

In the gearing systems, the excitation sources are the mesh stiffness fluctuations that are responsible for the system excitation.[10] In a stationary working regime, these fluctuations are periodic and associated with the mesh frequencies, which are the product of the number of teeth in one of the two gears and that gear's frequency of rotation.

Some precedent works assumed that the gears, shafts, and bearing are rigid and that the teeth contact could be modeled punctually by a spring along the line of action of the teeth contact. The stiffness amplitude also varies among the gear pairs.

This study regroups two gearing trains,[6,16] and at every train, the mesh stiffness variation can be modeled by a periodic wave function. Figure 3 illustrates this periodic fluctuation. The average value of the mesh stiffness is noted by k_{imoy}, which varies according to the type of gearing considered. Several numeric methods (including work with finite elements) have been developed to calculate this value.[12]

The extreme terms of the mesh stiffness are defined by:

$$k_{im} = -\frac{k_{imoy}}{2\times\varepsilon_{\alpha i}} \text{ and } k_{iM} = -k_{im} \times \frac{2-\varepsilon_{\alpha i}}{\varepsilon_{\alpha i}-1} \qquad \text{(Eq 2)}$$

where i designates the stage number (i = 1:2). The variable $\varepsilon_{\alpha i}$ represents the contact ratio at stage i. It is a geometric characteristic of the gears and generally takes a positive value between 1 and 2. However, Parker and Lin[12] showed that the integer values of $\varepsilon_{\alpha i}$ lead to weaker vibratory levels because, in this particular case, the number of couple teeth in contact remained constant during the work.

The two mesh frequencies are related to each other by:

$$fe_2 = \frac{Z_3}{Z_2}\times fe_1 \qquad \text{(Eq 3)}$$

where Z_2 and Z_3 represent the teeth numbers of gears 2 and 3.

A gearing pair always presents a backlash between the teeth to ensure lubrication and to eliminate interference and galling. Backlash is defined as the excess of thickness space of a tooth compared to the conjugated tooth thickness. The backlash value must not be exaggerated for the work requirements, but it should be sufficient to represent actual tooth-to-tooth contact loss.

The contact loss is modeled by the discontinuity of the defined meshing force following the line of action. Figure 4 shows this discontinuity according to the teeth deflection $\delta_i(t)$.[6]

The term $\tilde{f}_i(\delta_i(t), b_i)$ is a nonlinear symmetric function without zero value and is defined by:

$$\tilde{f}_i(\delta_i(t), b_i) = k_i(t)\tilde{h}_i(\delta_i(t), b_i) \qquad \text{(Eq 4)}$$

where b_i represents the backlash between two teeth before meshing, as shown in Fig. 4.

$\tilde{h}_i(\delta_i(t), b_i)$ is the displacement function defined on three domains:

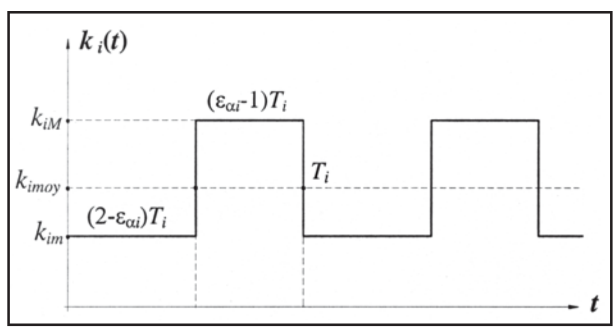

Fig. 3 Mesh stiffness fluctuation modeling

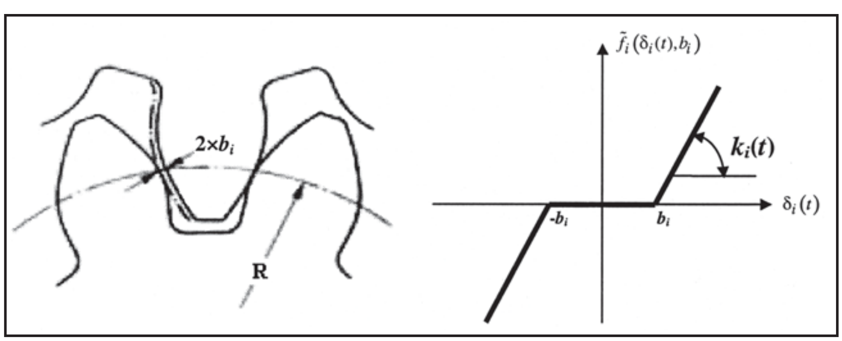

Fig. 4 Contact loss modeling

$$\tilde{h}_i(\delta_i(t), b_i) = \begin{cases} \delta_i(t) - b_i & \text{if} \quad \delta_i(t) > b_i \\ 0 & \text{if} \quad -b_i < \delta_i(t) < b_i \\ \delta_i(t) + b_i & \text{if} \quad \delta_i(t) < -b_i \end{cases}$$

(Eq 5)

Mathematically, this nonlinear function is modeled with some smoothening functions, $\tilde{g}_i(\delta_i(t) - b_i)$, as follows:[3]

$$\tilde{h}_i(\delta_i(t), b_i) = \delta_i(t) + \frac{\tilde{g}_i(\delta_i(t) - b_i) - \tilde{g}_i(\delta_i(t) + b_i)}{2}$$

(Eq 6)

The smoothening functions used are based on the work of Kim et al.[3] and include the hyperbolic-tangent type:

$$g_1(x(t)) = xtanh(\sigma x(t)) \quad \text{(Eq 7)}$$

and the arc-tangent type:

$$\tilde{g}_2(x(t)) = x\frac{2}{\pi}arctan(\sigma x(t)) \quad \text{(Eq 8)}$$

with $x(t) = \delta(t) \pm b$, where σ is a nonlinear parameter. Figure 5 presents the function $\tilde{h}_i(\delta_i(t), b_i = 4 \times 10^{-4})$ by using the two smoothening functions $\tilde{g}_1(\delta(t) \pm b)$ and $\tilde{g}_2(\delta(t) \pm b)$. A lower value of σ can reduce the calculating time and increase the simulation convergence, while an elevated value of σ (10^6) permits an approach to the original nonlinear function and is useful with the numeric simulation.

One of the criteria for the choice of a smoothing function is the method to integrate it into the theoretical model. This function must be compatible with the problem being solved.

In this paper, the nonlinear real behavior of a two-stage gear reducer is surveyed while supposing that the backlash is localized to the first train only. The second tooth contact is assumed to be perfect. The smoothening function used in this work is the hyperbolic-tangent type. Developing the first three terms of the Fourier-series expansion of the hyperbolic-tangent function, the function $\tilde{h}_i(\delta_i(t), b_i)$ can be formulated as follows:

$$\tilde{h}_i(\delta_i(t), b_i) = \tilde{A}_i(\delta_i(t), b_i)\delta_i(t) \quad \text{(Eq 9)}$$

with

$$\tilde{A}_i(\delta_i(t), b_i) = A_{1i} + A_{2i}\delta_i^2(t);$$

$$A_{1i} = 1 - 2\sigma_i b_i + \frac{4}{3}\sigma_i^3 b_i^3$$

and $A_{2i} = \frac{4}{3}\sigma_i^3 b_i$.

Nonlinear Problem Formulation

The Lagrange formalism was used to formulate the equation of movement of the two-stage gear reducer to every degree of freedom $\delta_i(t)$ of the system. The general matrix form of this two-degrees-of-freedom nonlinear system is defined by:

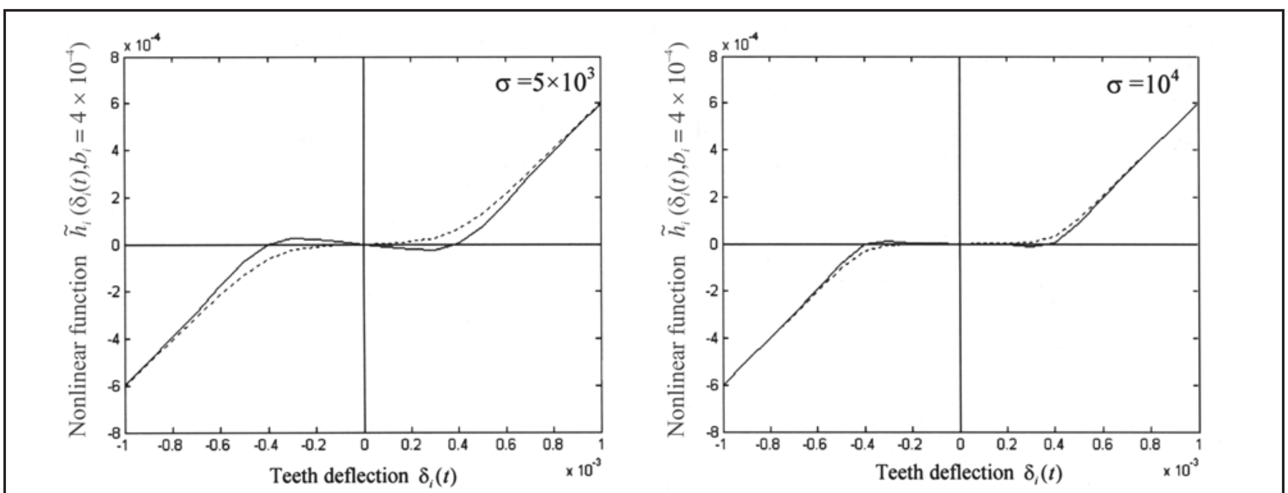

Fig. 5 *Nonlinear function $\tilde{h}_i(\delta_i(t), b_i = 4 \times 10^{-4})$ using smoothening functions $\tilde{g}_1(\delta(t) \pm b)$, indicated by solid line, and $\tilde{g}_2(\delta(t) \pm b)$, indicated by dotted line*

$$[M]\begin{Bmatrix}\ddot{\delta}_1(t)\\ \ddot{\delta}_2(t)\end{Bmatrix}+[C]\begin{Bmatrix}\dot{\delta}_1(t)\\ \dot{\delta}_2(t)\end{Bmatrix}+[K(t)]\{\tilde{H}(\delta_i(t),b_i)\}=\{F\} \quad \text{(Eq 10)}$$

where M is the system mass matrix, defined as:

$$[M]=\begin{bmatrix}m_1 & 0\\ 0 & m_3\end{bmatrix} \quad \text{(Eq 11)}$$

and $K(t)$ represents the system mesh stiffness matrix, defined as:

$$[K(t)]=\begin{bmatrix}k_1(t) & \dfrac{m_1}{m_2}k_2(t)\\ \dfrac{m_3}{m_2}k_1(t) & k_2(t)\end{bmatrix} \quad \text{(Eq 12)}$$

The term m_i represents the equivalent masses of the system:

$$\frac{1}{m_1}=\left(\frac{r_1^2}{I_1}+\frac{r_2^2}{I_2+I_3}\right);\ \frac{1}{m_2}=\left(\frac{r_2r_3}{I_2+I_3}\right) \text{ and}$$

$$\frac{1}{m_3}=\left(\frac{r_2\,r_3}{I_2+I_3}+\frac{r_4^2}{I_4}\right) \quad \text{(Eq 13)}$$

and r_j and I_j represent the base radius and the inertia moments, respectively, of the gears j (j = 1…4).

In most previous works, a viscous damping has been considered to reduce the level of vibrations. The expression of the equivalent viscous damping (Rayleigh damping) is given by:

$$[C] = \mu\,[M] + \kappa\,[K_{moy}] \quad \text{(Eq 14)}$$

where μ and κ are the damping coefficients, and K_{moy} is the average mesh stiffness matrix of the system.

The displacement vector that introduces the nonlinear character in the model is defined by:

$$\{\tilde{H}(\delta_i(t),b_i)\}=\begin{Bmatrix}\tilde{h}_1(\delta_1(t),b_1)\\ \delta_2(t)\end{Bmatrix} \quad \text{(Eq 15)}$$

When using Eq 15, the two-degrees-of-freedom nonlinear differential equation of the system will be given by the following form:

$$[M]\begin{Bmatrix}\ddot{\delta}_1(t)\\ \ddot{\delta}_2(t)\end{Bmatrix}+[C]\begin{Bmatrix}\dot{\delta}_1(t)\\ \dot{\delta}_2(t)\end{Bmatrix}+[K(t)] \quad \text{(Eq 16)}$$
$$\begin{bmatrix}\tilde{A}_1(\delta_1(t),b_1) & 0\\ 0 & 1\end{bmatrix}\begin{Bmatrix}\delta_1(t)\\ \delta_2(t)\end{Bmatrix}=\{F\}$$

The excitation sources of the system are the periodic fluctuations of stiffness of the two meshes. To introduce the excitation term (mesh frequency) in the equation, the following variable change is used:

$$\tau = \omega_1 t \quad \text{(Eq 17)}$$

where ω_1 and τ are the first mesh frequency excitation and the new dimensionless variable of the system, respectively. In accordance with this dimensionless variable, the mesh stiffness fluctuation is represented in Fig. 6.

Numeric Responses

The effect of the first mesh frequency on the nonlinear dynamic responses of the system is found by solving the following two-degrees-of-freedom nonlinear differential equation according to the dimensionless variable τ:

$$\omega_1^2\,[M]\begin{Bmatrix}\ddot{\delta}_1(\tau)\\ \ddot{\delta}_2(\tau)\end{Bmatrix}+\omega_1\,[C]\begin{Bmatrix}\dot{\delta}_1(\tau)\\ \dot{\delta}_2(\tau)\end{Bmatrix}+[K(\tau)]$$
$$\begin{bmatrix}\tilde{A}_1(\delta_1(\tau),b_1) & 0\\ 0 & 1\end{bmatrix}\begin{Bmatrix}\delta_1(\tau)\\ \delta_2(\tau)\end{Bmatrix}=\{F\} \quad \text{(Eq 18)}$$

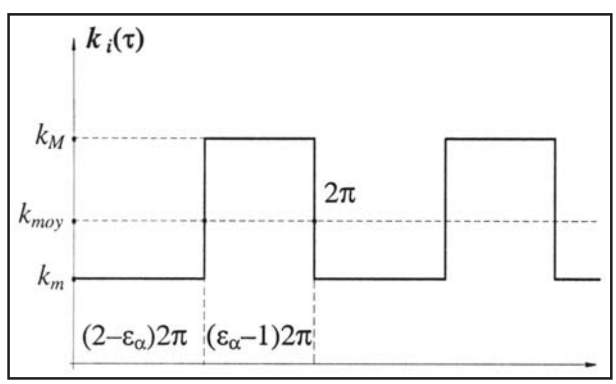

Fig. 6 *Mesh stiffness fluctuation $k_i(\tau)$ according to the dimensionless variable τ*

The nonlinear solution of this differential equation is obtained through a step-by-step numerical iteration method by using the Newmark algorithm.[17] This approach permits a fast resolution of the problem for a chosen mesh frequency ω_1. Indeed, for every step $\Delta\tau$, the two-degrees-of-freedom nonlinear differential equation is written in the general following form:

$$G(X(t)) = \left[G_1(X(t)); G_2(X(t))\right] = 0 \qquad \text{(Eq 19)}$$

where $X(t)$ is a vector formed by two deflections $\delta_i(t)$. The Newton-Raphson algorithm consists of substituting each equation $G_i\ (X(t)) = 0$ with its first development order from the previous evaluation. This method is based on the minimization of the mathematical difference of the system responses to the iterations $k + 1$ and k for the same mesh frequency ω_1.

The resolution of the motion equation (Eq 18) is obtained through a step-by-step iterative method for a chosen mesh frequency while using the Newmark method, which permits a direct integration. The nonlinear problem is solved by the Newton-Raphson method. This approach causes the solutions to converge quickly, and, with reasonable initial assumptions, it can normally find a root after only three or four iterations. Several studies showed that this method is often used to solve nonlinear functions, and it avoids the calculation of the nonlinear transitions. For gearing systems with the introduced nonlinearity, the Newton-Raphson method is compatible with the smoothening functions. Also, the lower σ values should ensure minimal numeric difficulties when solving the nonlinear differential equations using the Newton-Raphson method. The computing procedure for the nonlinear response is described by the schematic shown in Fig. 7.

Figure 8 represents the temporal fluctuations of the teeth deflection for the first mesh frequency ω_1 = 2000 rad/s. The spectrum associated with these fluctuations shows peaks at the first and second mesh

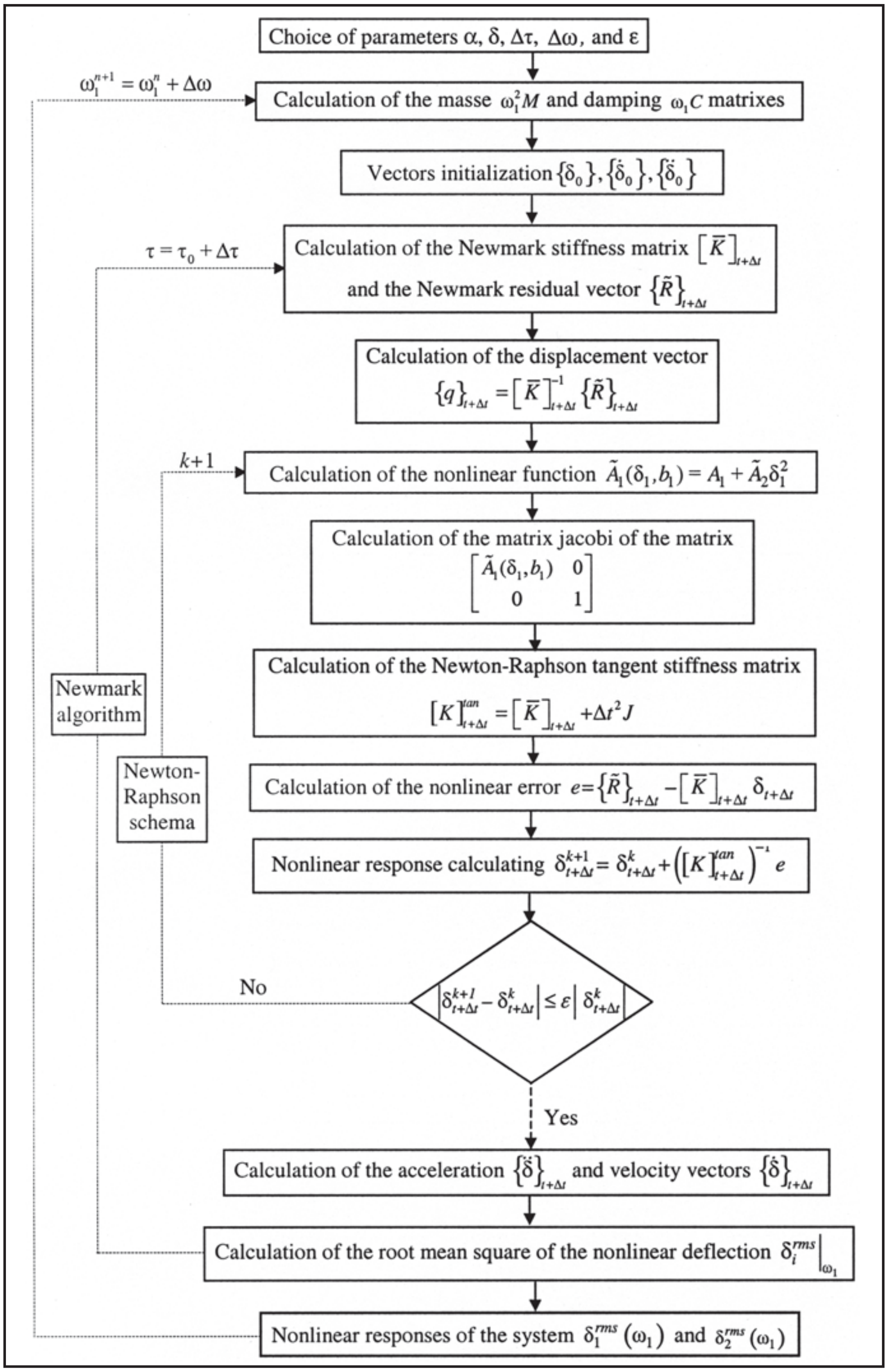

Fig. 7 *Flow diagram of the Newmark and Newton-Raphson algorithms*

frequencies and their first harmonics. Thus, the system is excited only by the periodic fluctuations of the mesh stiffness.

The nonlinearity introduced in the structure clearly appears while passing by a sweep on the first mesh frequency ω_1. Figure 9 represents the root

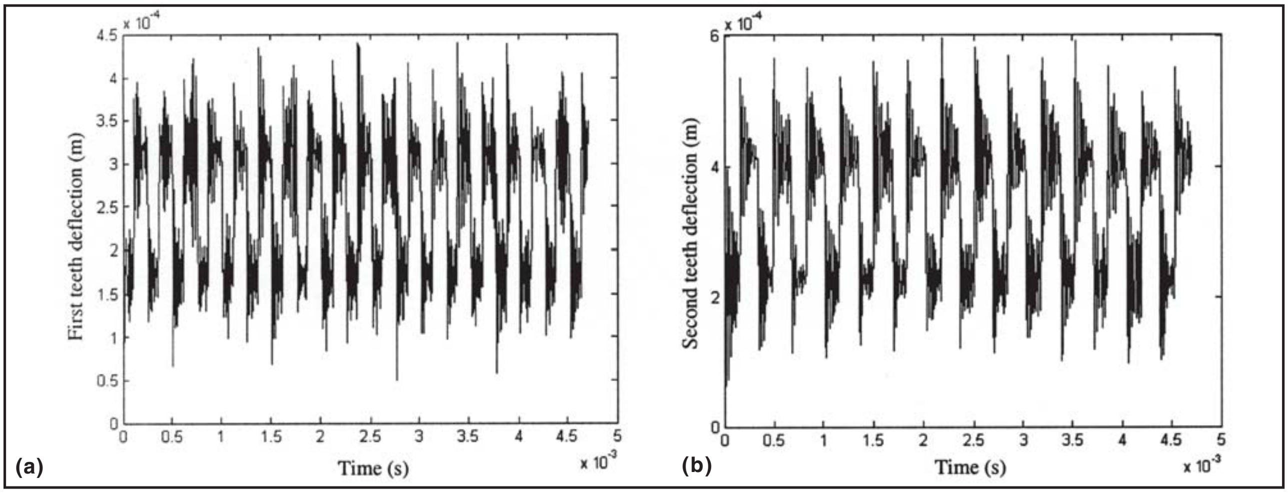

Fig. 8 *Nonlinear temporal responses for* ω_1 = 2000 rad/s. $\mu = 10^{-2}$, $\kappa = 10^{-6}$. (a) $\delta_1(t)$. (b) $\delta_2(t)$

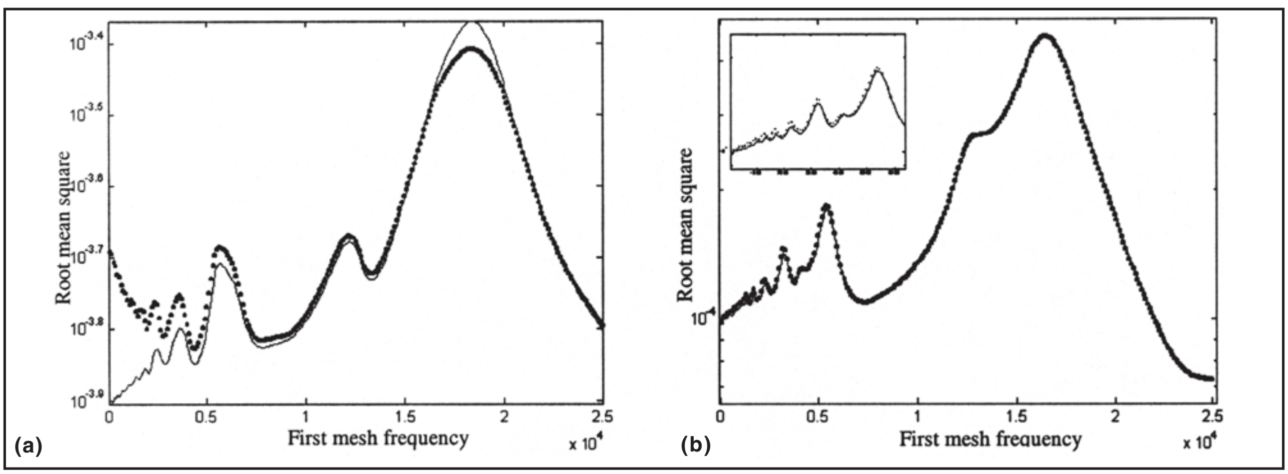

Fig. 9 *Frequency response of the system.* $\mu = 10^{-2}$, $\kappa = 10^{-6}$. (a) $\delta_1(t)$. (b) $\delta_2(t)$. *Solid line, linear behavior; dotted line, nonlinear behavior for* $b_1 = 510^{-4}$; $\sigma_1 = 2000$

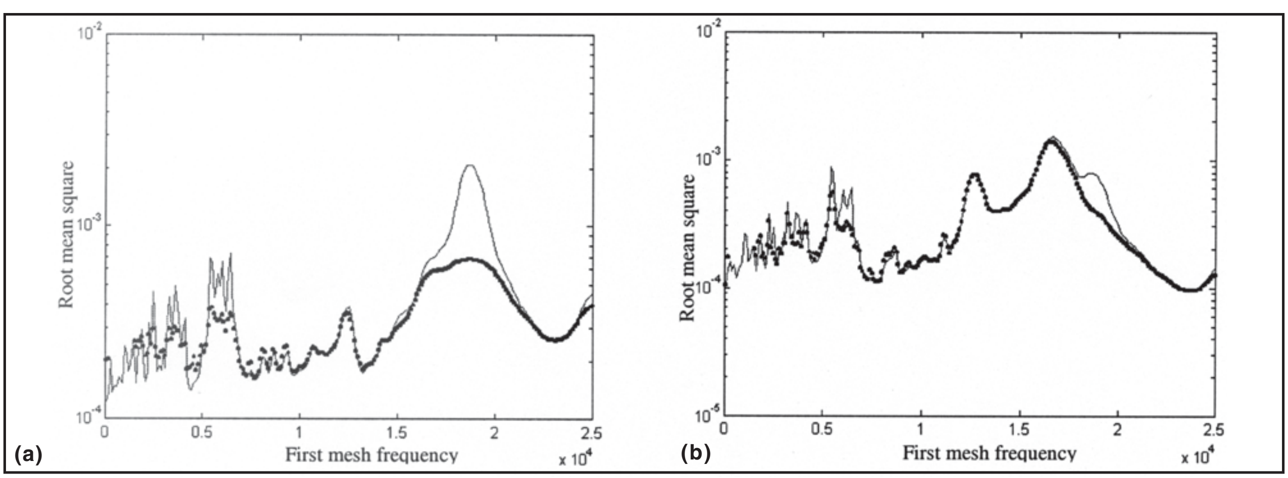

Fig. 10 *Effect of damping coefficients on the frequency response* $\mu = 10^{-2}$, $\kappa = 10^{-7}$. (a) $\delta_1(t)$. (b) $\delta_2(t)$. S*olid line, linear behavior; dotted line, nonlinear behavior for* $b_1 = 510^{-4}$; $\sigma_1 = 2000$

mean square variation of the teeth deflections according to ω_1 for the case of a linear and nonlinear system. For weak speeds, the nonlinearity appears clearly when the root mean square variation of the first tooth deflection is presented, because the backlash is introduced only on the first stage. The critical speeds of rotation correspond to the eigen values of the structure ($\omega' = 1.2 \times 10^4$ rad/s and $\omega'' = 1.8 \times 10^4$ rad/s) and its first harmonics.

Figure 10 represents the effect of the damping coefficients μ and κ on the system frequency responses. The appearance of other peaks corresponding to the other harmonics of the eigen values of the system is noted. The nonlinear effect appears more and more clearly around the second eigen frequency, $\omega'' = 1.8 \times 10^4$ rad/s.

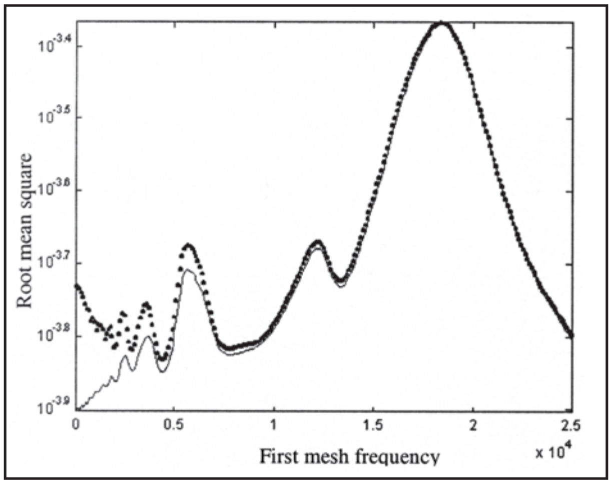

Fig. 11 *Effect of backlash value on the frequency response* $\delta_1(t)$. $b_1 = 10^{-4}$; $\sigma_1 = 2000$; $\mu = 10^{-2}$, $\kappa = 10^{-6}$

To study the backlash effects, the value of the "gap" shown in Fig. 4 was changed. Figure 11 shows that the effect of nonlinearity is reduced for weak speeds and will be attenuated around ω'' for a weak value of backlash.

Figure 12 shows the effect of the integration step $\Delta\tau$ on the frequency responses associated with the first deflection. The integration step $\Delta\tau$ must be thinner so that the algorithm can adapt to the very abrupt changes in the nonlinear function $\tilde{A}(\delta(t), b)$.

Conclusions

A new gear dynamic model is proposed to incorporate the nonlinearities induced by backlash. The torsional model presented in this paper introduces the teeth flexibilities in the two-stage gear system affected by a localized nonlinear stiffness function caused by tooth loss during service. This model is compared with a linear time-varying system formulation and the nonlinear system; the Newton-Raphson algorithm is applied, with backlash acting as an internal excitation. The system is more sensitive to errors in the distance between teeth than to fluctuations in the stiffness magnitude, although the qualitative effect is similar. Several parameters affect the nonlinear behavior of the system, such as the nonlinearity coefficients b_1 and σ_1, the damping coefficients μ and κ, and the integration step $\Delta\tau$.

An important component of this work consists of studying the nonlinear modal analysis of the model. This part is characterized by the complexity coming

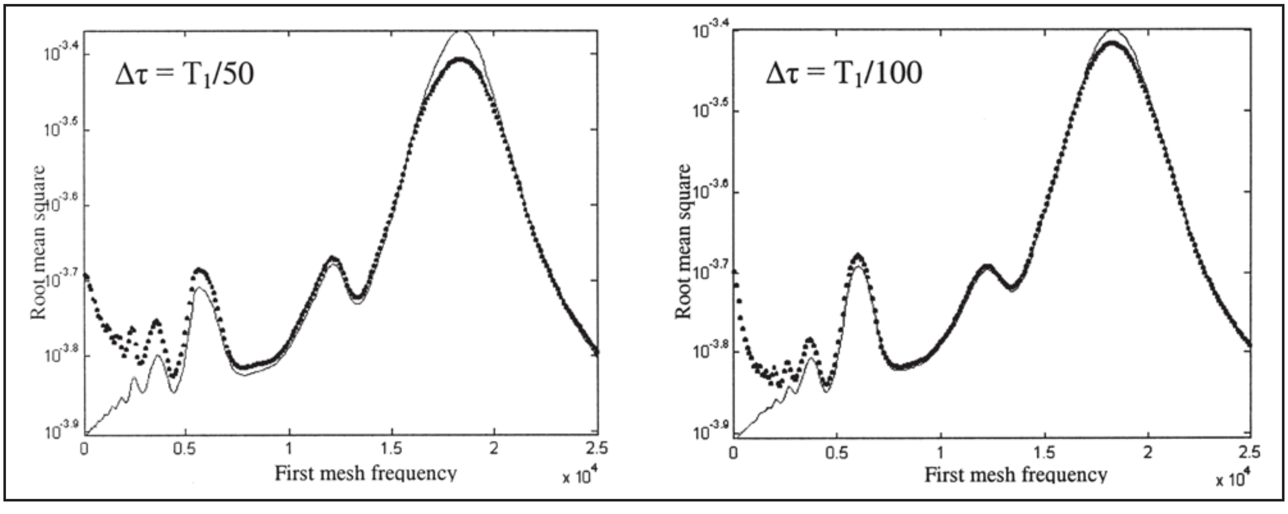

Fig. 12 *Effect of integration step* $\Delta\tau$ *on the frequency response* $\delta_1(t)$. $b_1 = 510^{-4}$; $\sigma_1 = 2000$; $\mu = 10^{-2}$, $\kappa = 10^{-6}$

from the periodic fluctuations of the mesh stiffness and will be the object of future investigations. However, this study clearly shows that backlash impacts the vibrational frequency of a gear system and therefore may impact the performance of the machine.

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